A high-tech, export-driven economy could make Bangladesh the next Flying Geese follower.

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Introduction

The Flying Geese Paradigm is an economic model introduced by the economist Kaname Akamatsu. It describes how developing nations, especially in East Asia, industrialize rapidly by catching up with more advanced economies. Akamatsu showed how Japan, the lead goose, passed basic manufacturing and electronic assembly down to the first-tier followers (South Korea, Taiwan, Hong Kong and Singapore), who in turn passed it on to second-tier nations such as Malaysia, Thailand and Indonesia.

Akamatsu's original study did not mention Vietnam. We include it here because its economic path closely resembles that of Bangladesh, which is our focal point. Each of these countries followed the same rule of thumb: Import–Production–Export. They imported advanced goods, began manufacturing domestic substitutes, and eventually moved their economies toward an export-driven trade pattern in high-tech industries worldwide.

Source of Data

The secondary data come from the World Development Indicators (WDI) of the World Bank. They cover nine countries (Bangladesh, China, Indonesia, Japan, Korea Rep., Malaysia, Philippines, Singapore and Vietnam) and 24 indicators, including GDP per capita, Manufacturing value added (% of GDP) and High-technology exports (% of manufactured exports). The sample period runs from 1975 to 2025.

Methodology

To keep our conclusions statistically and econometrically reliable, the study combines several time-series comparisons with an extensive literature review. The dependent variable is GDP per capita growth (annual %). We want to observe how Bangladesh behaves over time compared with other large and growing economies. To identify structural breaks in annual fluctuations, we used the Chow test, which detects changes in economic behavior across years. To locate the breaking points themselves, we also used the Zivot-Andrews endogenous structural break unit root test.

For a clearer comparison among countries, we used the following:

1. **Chow Test and Zivot-Andrews Test:** structural breaks in economic behavior, and the years in which they occur.
2. **Seasonality Decomposition Method:** tests whether genuine annual growth is visible once trend, seasonal and residual components are separated.
3. **Spearman Rank Correlation:** compares Bangladesh with the leading geese in terms of manufacturing and high-technology exports.
4. **Net Trade Competitiveness Ratio:** helps us understand endogenous growth.
5. **Kojima FDI-Upgrading Quadrant:** plots Foreign Direct Investment against high-technology exports over the decades to show each country's economic behavior and trajectory.
6. **Granger Causality Network:** a stochastic graph that shows directly how different interconnected indicators contribute to growth.
7. **Augmented Dickey-Fuller and KPSS Tests:** check the stationarity (unit root behavior) of the indicators for each country, especially Bangladesh.

Workflow

The rest of the study follows the workflow below and shows how a high-tech, export-based economy could be the key to Bangladesh's prosperity.

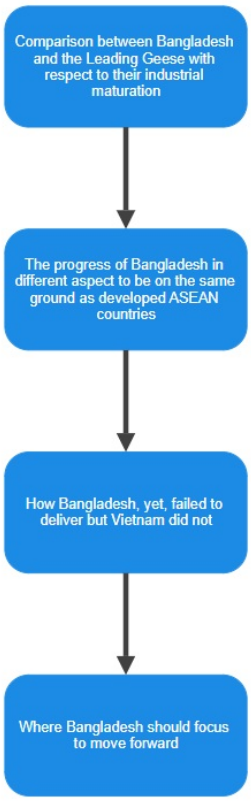


Figure 1. Workflow of the study

Comparison with Bangladesh

According to our WDI analysis, Bangladesh's current economic progress looks stagnant and flat, despite a severe drop in GDP per capita growth after COVID-19. However, the Chow test results show that Japan, China, Korea Rep. and Singapore did not suffer a pandemic-era decline. So before we discuss growth per capita, the pandemic needs to be addressed. The Chow test results for the given panel regression are as follows:

Country Name	Breakpoint Years
Bangladesh	2019
China	N/A
Indonesia	1995–2019 (every year)
Japan	N/A
Korea, Rep.	1995–2008 (every year)
Malaysia	1999, 2000, 2005, 2013–2019
Philippines	2016–2019
Singapore	N/A
Viet Nam	2019

Almost every country was economically affected by the COVID-19 pandemic, except China, Singapore, South Korea and Japan, which we know as the leading geese.

With that acknowledged, we can look at Bangladesh's situation through the decomposed time-series graph of GDP per capita growth (% annual).

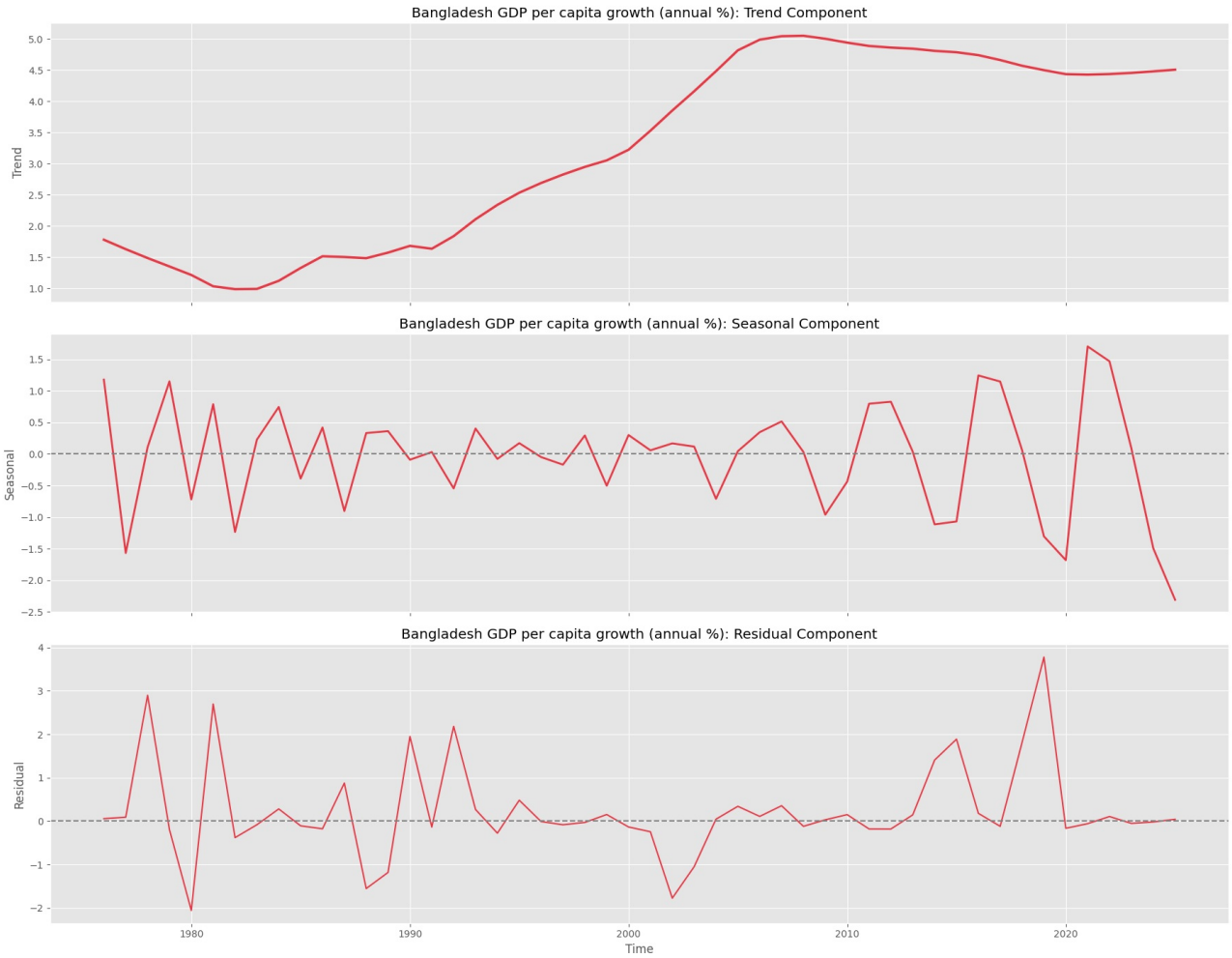


Figure 2. Bangladesh GDP per capita growth (annual %): trend, seasonal and residual components

The graph shows that the seasonality of the Bangladesh economy is very inconsistent, but before 2019 the trend was moving upward.

To compare Bangladesh with these leading countries, we looked at the similarity of their economic strategies, first among the giants and then with Bangladesh. The Spearman rank correlation heatmap below gives a clear picture of the relationship between the manufacturing industry and high-tech exports across countries, at lags from the current year up to 10 years:

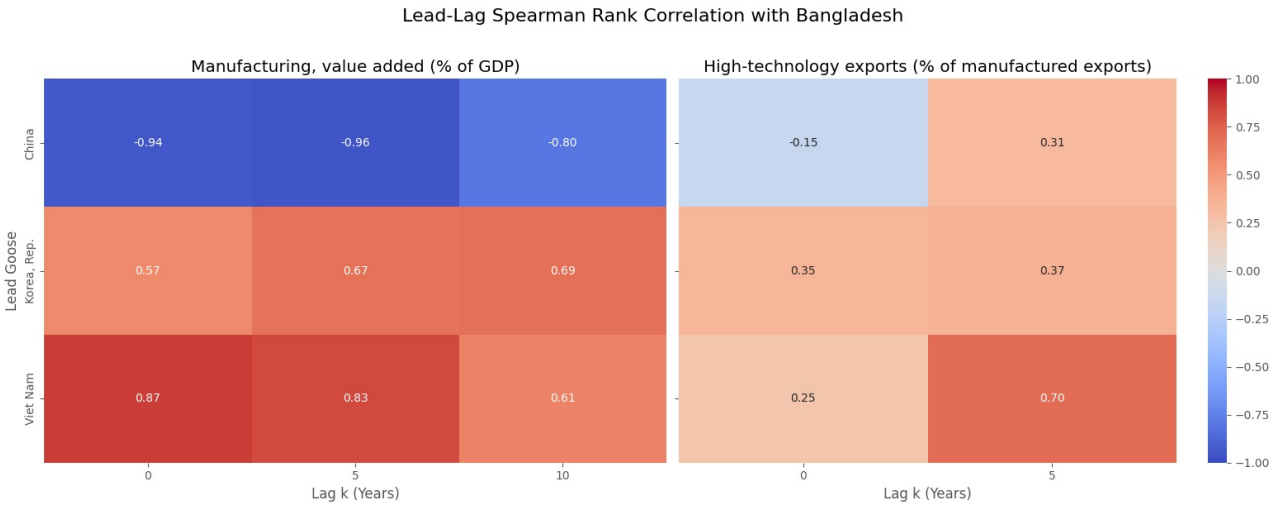


Figure 3. Lead-lag Spearman rank correlation with Bangladesh

Bangladesh depends heavily on the RMG sector, so it makes sense that Vietnam is highly and positively correlated with it in manufacturing over the last five years; a decade earlier the correlation was still positive but only moderate. South Korea shows a moderate correlation with Bangladesh in manufacturing. The most interesting result is that China shows a strong *negative* correlation with Bangladesh in the same field, which is notable because China is one of Bangladesh's main competitors in the global RMG market.

For high-technology exports (% of manufactured exports), Bangladesh shows a weak correlation with almost all of these countries, the exception being Vietnam. The strong correlation at lag 5 exists because Vietnam only recently started producing high-technology goods and earning profitable exports from them, which was not the case five years earlier.

Pacing of Bangladesh

Before COVID-19, Bangladesh's GDP per capita growth was trending upward. But an upward trend does not show the whole picture. To follow the Import–Production–Export pattern, we focused on export- and import-related variables such as Net Trade Competitiveness (NTC), defined as:

> **NTC = (Exports (% of GDP) − Imports (% of GDP)) / Trade (% of GDP)**, calculated for each country i and year t.

NTC gives a clear view of a country's trade history over time. We compared the countries with Bangladesh in the following graph to see the gap between exports and imports:

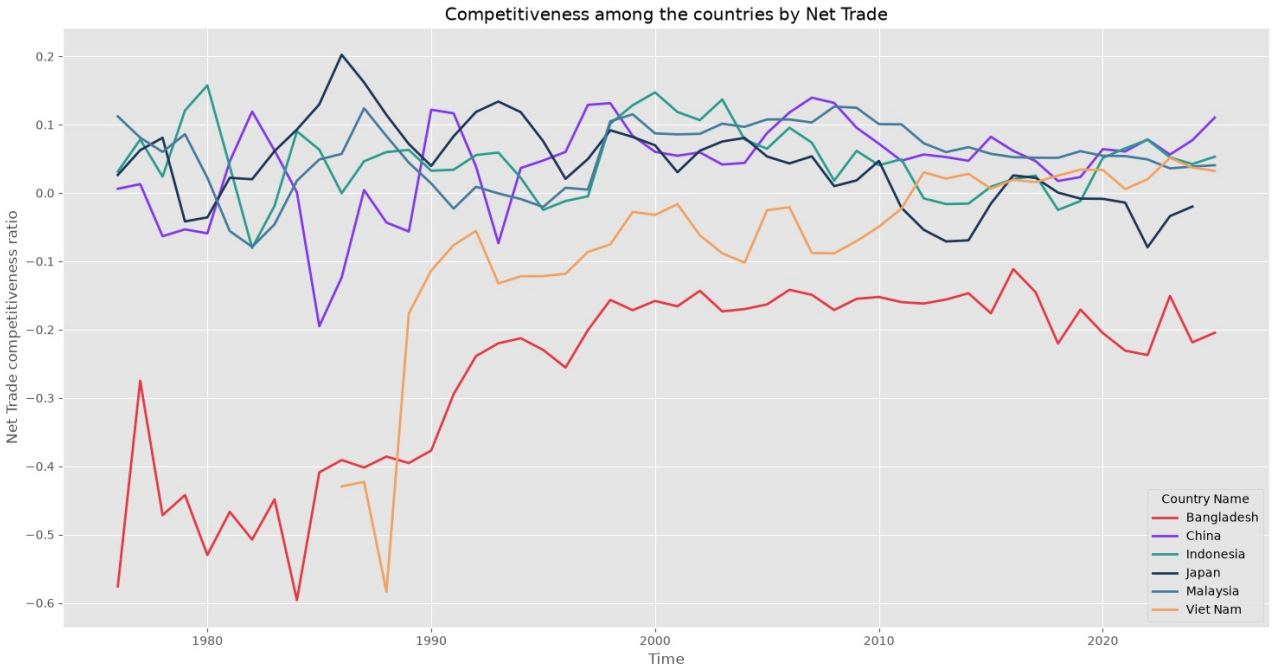


Figure 4. Competitiveness among the countries by Net Trade

Most of the countries we considered have a positive NTC. In the 1980s Vietnam's NTC curve was below Bangladesh's, but after 2012 it crossed the zero line and kept growing. Bangladesh, however, stays below the zero line throughout. Its trade balance is always negative, unlike the developed and developing countries, both the flying geese leaders and the followers.

Could this be because Bangladesh's economy is more inward-moving, meaning it depends mostly on Gross Capital Formation rather than Foreign Direct Investment (FDI)? Here is the trajectory of Gross Capital Formation for these countries over the past decades:

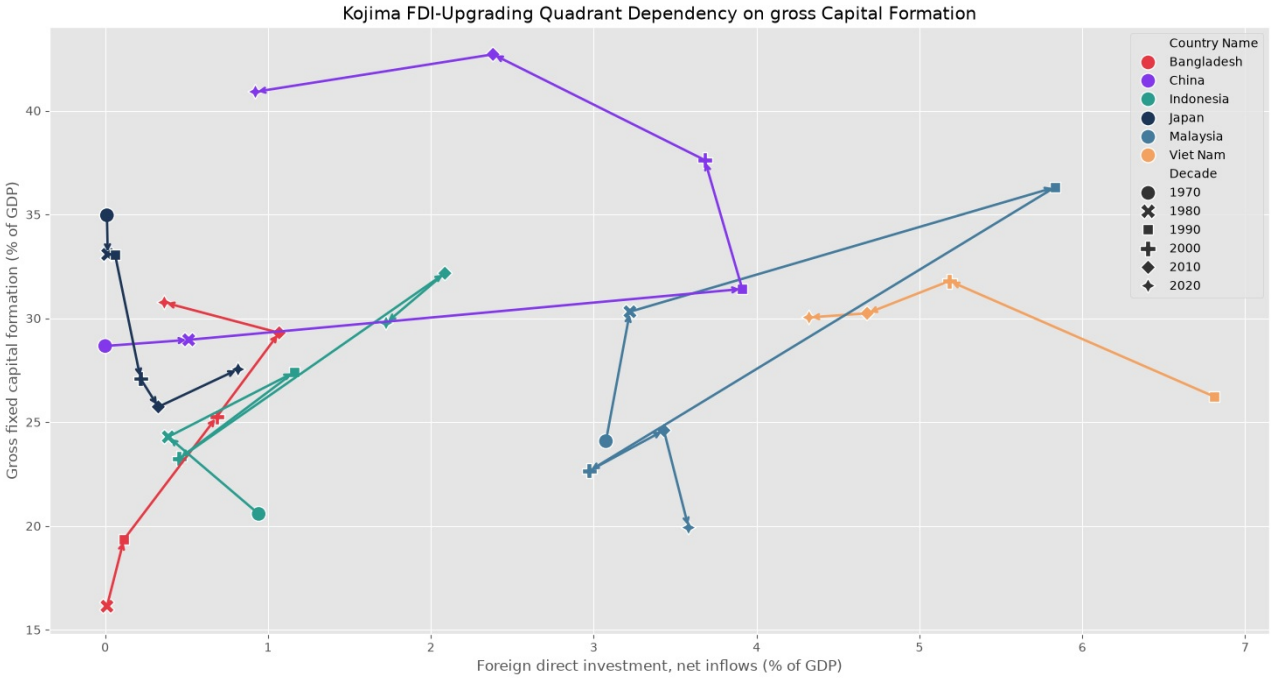


Figure 5. Kojima FDI-upgrading quadrant: dependency on gross capital formation

According to the Kojima FDI-upgrading quadrant, Bangladesh has been heavily dependent on an inward-moving economy, relying on gross capital formation, for the past few decades. Many countries show a rising dependence on gross capital formation, while some show a declining trajectory when it comes to FDI.

Is Bangladesh struggling because of its massive population, in terms of employment, health, female empowerment and other indicators? To find out, we compare Bangladesh with Vietnam, which recently crossed the zero NTC margin, and check whether population growth correlates with the strong growth of Vietnam's manufacturing.

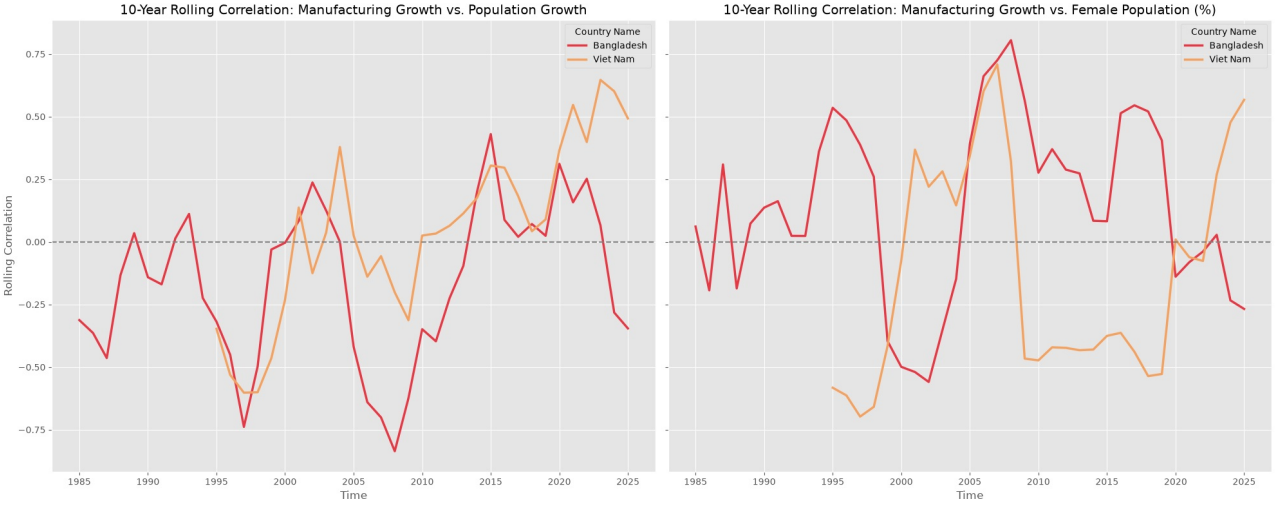


Figure 6. 10-year rolling correlation of manufacturing growth with population growth and female population share

Female employment should be taken into consideration, since employing half of the population is an important factor for economic growth.

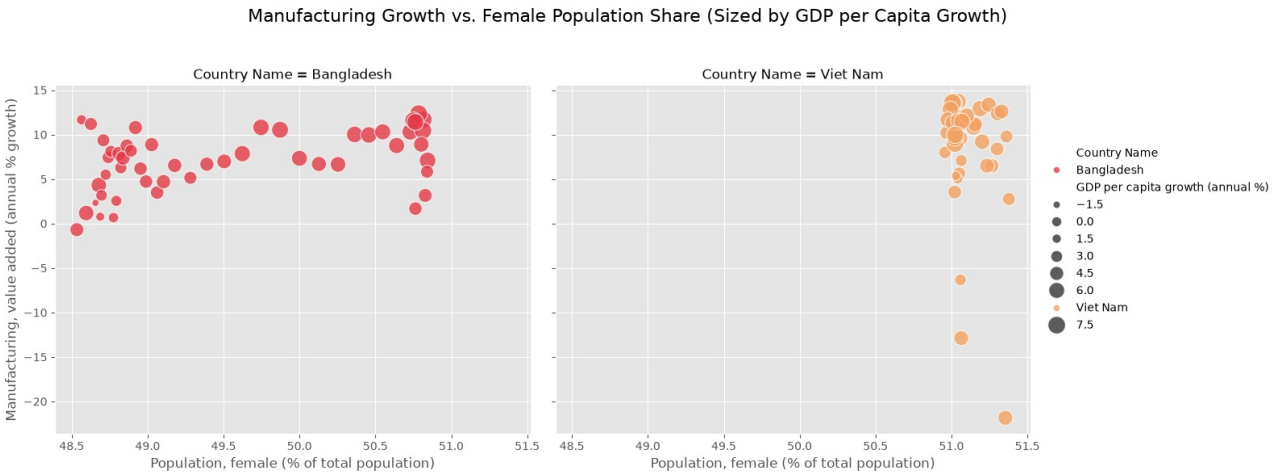


Figure 7. Manufacturing growth vs. female population share (sized by GDP per capita growth)

The scatterplot shows that the majority of Vietnam's population is female, and that Vietnam has managed to bring a large share of women into employment, supporting both manufacturing growth and GDP growth. This is another area where Bangladesh can improve.

Why Bangladesh Is Failing

Akamatsu showed that follower countries focused on globally trending manufactured products, mostly assembling imported raw materials and then exporting them in bulk for profit. In recent decades, high-tech industries have been booming. Technology keeps advancing and the demand for digital products has risen rapidly. Both leading and follower countries adapted their industries to the global market alongside their local market. This not only employed their populations but also raised manufacturing's contribution to GDP and helped them keep a stable NTC above the zero line. So let's look at Bangladesh's movement in the Kojima FDI-upgrading quadrant against the high-tech share of manufactured exports:

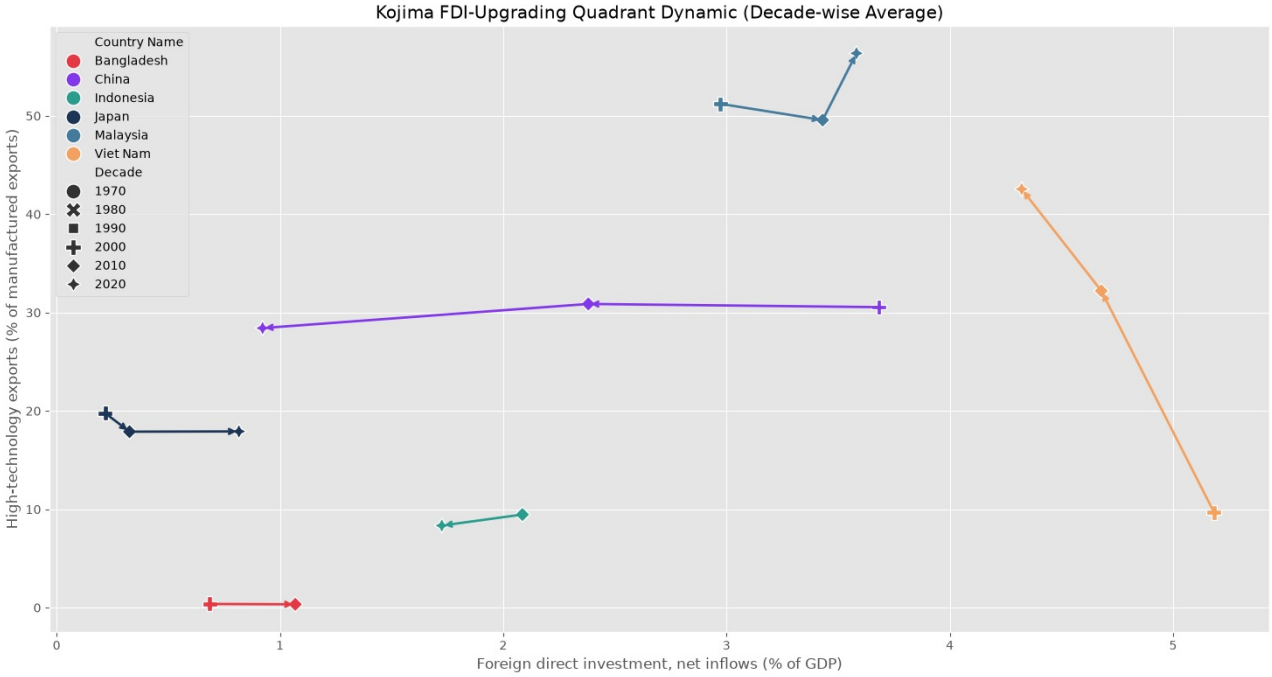


Figure 8. Kojima FDI-upgrading quadrant dynamics (decade-wise average)

Because Vietnam attracted high FDI (4–5% of GDP), its high-tech exports rocketed from **10% in the 2000s to over 42% in the 2020s**. Bangladesh, starved of FDI, has **flatlined at around 0.5% high-tech exports**. This explains why Bangladesh has not kept pace as a follower country, while Vietnam, on the other side of

the coin, achieved great and consistent success.

The Structural Break & Broken Causality

Bangladesh has not been able to create any substantial growth in GDP per capita. The Zivot-Andrews endogenous break tests show several breaks, each tied to a different part of Bangladesh's economic context:

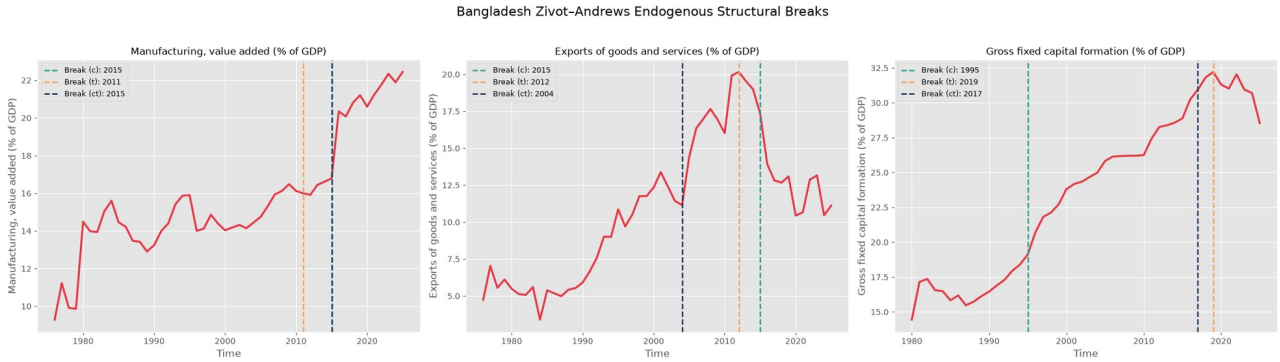


Figure 9. Bangladesh Zivot-Andrews endogenous structural breaks

Manufacturing value added (% of GDP) had a huge spike in 2015. Exports of goods and services (% of GDP) broke in 2004, raising the share from 11.25% to as much as 20.00%. Gross fixed capital formation (% of GDP) spiked more recently, in the 2018–2019 period. *In simple words:* after 2012–2015, Bangladesh's manufacturing growth became increasingly inward-looking (serving the domestic market rather than expanding its export share), while private capital formation began losing steam.

Next, the Dumitrescu-Hurlin and Toda-Yamamoto causality tests show why Bangladesh failed to upscale its market. The causality network graph below compares the regional picture with Bangladesh, to show where exactly Bangladesh should focus its effort instead of depending only on local markets, an inward-moving economy and domestic resources.

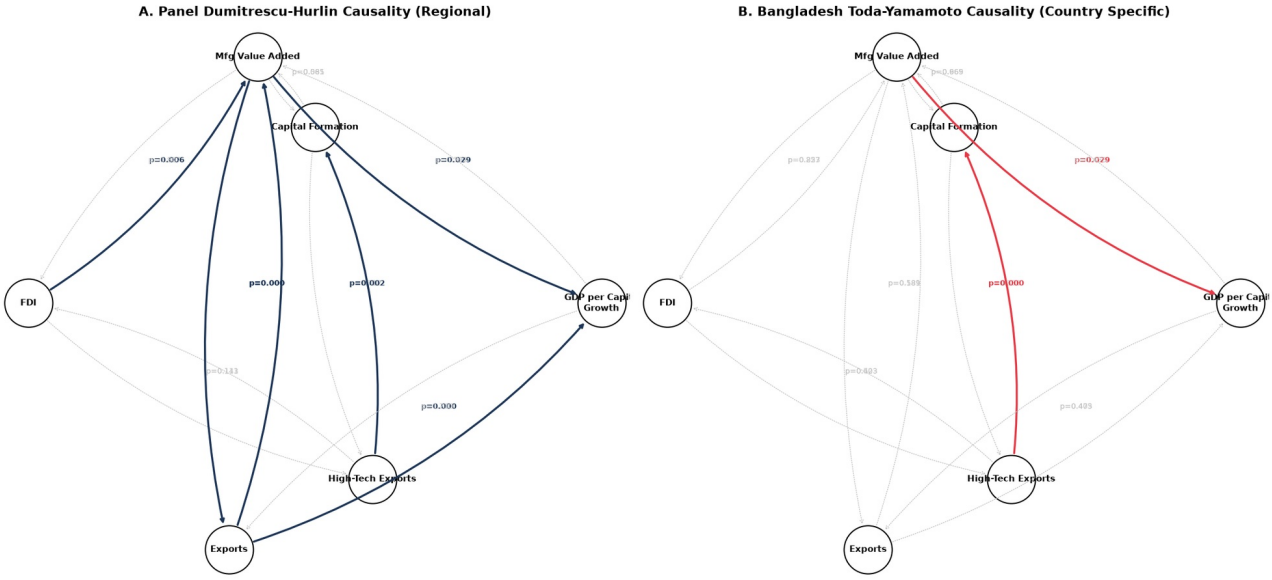


Figure 10. Causality network: (A) panel Dumitrescu-Hurlin (regional) and (B) Bangladesh Toda-Yamamoto (country specific)

FDI strongly affects manufacturing value added for the East Asian regional countries (statistically significant, p = 0.006), whereas the same link for Bangladesh fails (statistically insignificant, p = 0.853). Because Bangladesh's exports collapsed in 2012 and the economy relies heavily on imports, the link between manufacturing and exports also fails for Bangladesh (p = 0.589), though it holds for the other ASEAN countries. By strengthening its manufacturing industries, Bangladesh can maximize GDP growth,

given the causal connection between these two nodes ($p = 0.029$).

More interestingly, the high-tech exports link ($p = 0.000$, below 0.05) is also very helpful for Bangladesh. But we saw earlier that Bangladesh's high-tech exports are flatlined at around 0.5%. This redefines Bangladesh's objectives: it needs policy changes that ignite high-tech exports, which would directly stimulate a new wave of capital formation.

Results and Discussion

The decomposition of GDP growth reveals that, as the mature East Asian economies decelerated, Bangladesh and Vietnam emerged as the fastest-growing economies in the panel, sustaining trend growth rates near 6% to 7%. This acceleration aligns closely with Akamatsu's inter-country structural shift. Lead-lag Spearman rank correlations confirm the "baton pass" of labor-intensive manufacturing: China's declining manufacturing value-added share shows a near-perfect inverse correlation with Bangladesh's rising share ($\rho = -0.94$ at lag 0; $\rho = -0.96$ at lag 5). Bangladesh therefore absorbed the first wave of regional industrialization alongside Vietnam ($\rho = +0.87$), and its cyclical growth was insulated from external financial shocks by the inelastic global demand for its baseline apparel exports.

Unlike the standard Kojima hypothesis, which holds that Foreign Direct Investment (FDI) drives industrial upgrading, Bangladesh financed its first-wave expansion almost entirely through domestic channels. While peer economies like Malaysia and Vietnam kept FDI inflows between 3% and 7% of GDP to fund their structural transitions, Bangladesh's FDI stayed stagnant below 1%. Instead, Bangladesh substituted foreign capital with aggressive domestic investment, driving Gross Fixed Capital Formation from 16% to over 30% of GDP. Alongside this capital accumulation, demographic shifts supplied the labor flexibility it needed: a 10-year rolling correlation shows that Bangladesh's manufacturing expansion moved closely with the rising female share of the population, which grew from 48.5% to nearly 51%.

Despite mastering the first wave, Bangladesh shows a severe structural stall in the "second wave" of capital- and skill-intensive upgrading. Decade averages in the Kojima quadrant show Vietnam using its high FDI to pivot from 10% high-technology exports in the 2000s to over 42% in the 2020s. Bangladesh, by contrast, was starved of the technological spillovers that come with "pro-trade FDI" and remains flatlined at about 0.5% high-technology exports. This failure to move up into intermediate inputs and capital goods leaves it chronically reliant on imported manufacturing components. Consequently, while every other regional peer moved into a net-exporter position over its development cycle, Bangladesh's aggregate net trade competitiveness ratio hit a permanent ceiling around -0.15, trapping the economy in a structural trade deficit.

The Zivot-Andrews endogenous structural break tests pin down the timeline of this stall. Manufacturing value added experienced a positive regime shift in 2015, but exports as a share of GDP suffered a sharp downward break after 2012, collapsing from a peak of 20% back toward 11%. This points to a pivot toward inward-looking, consumption-driven growth instead of continued export-led industrialization. The comparative Granger-causality networks support this truncated mechanism. Across the broader Asian panel, the Dumitrescu-Hurlin test confirms robust causality from FDI to manufacturing ($p = 0.006$) and a bidirectional feedback loop between manufacturing and exports ($p < 0.001$). Within Bangladesh, the Toda-Yamamoto country-specific tests show that both channels are statistically severed ($p = 0.853$ and $p = 0.589$, respectively). The only transmission channels still active in Bangladesh run from manufacturing to GDP growth ($p = 0.029$) and from high-tech exports to capital formation ($p < 0.001$), which suggests that future capital expansion depends entirely on diversifying into higher-complexity exports.

Bottomline

Bangladesh pulled off an economic miracle, but it built it with a completely different engine from the rest of Asia. When China became too expensive to make the world's clothes, Bangladesh caught that manufacturing wave. But while countries like Vietnam used foreign investment (FDI) to buy new technology and move up from making t-shirts to making microchips, Bangladesh funded its own factory floor. It relied on the savings

of local entrepreneurs (domestic capital) and the sheer willpower of a massive, newly mobilized female workforce.

This home-grown approach worked brilliantly for 30 years, but the data shows the engine is now maxed out. Because Bangladesh never brought in the foreign tech investors who could have taught it to make complex machinery or electronics, it remains stuck at the very bottom of the high-tech export ladder. As a result, even though it exports billions of dollars in garments, it still has to buy its raw materials, chemicals and sewing machines from abroad, which keeps the country in a permanent trade deficit. The statistical tests show that the export engine effectively stalled around 2012–2015.